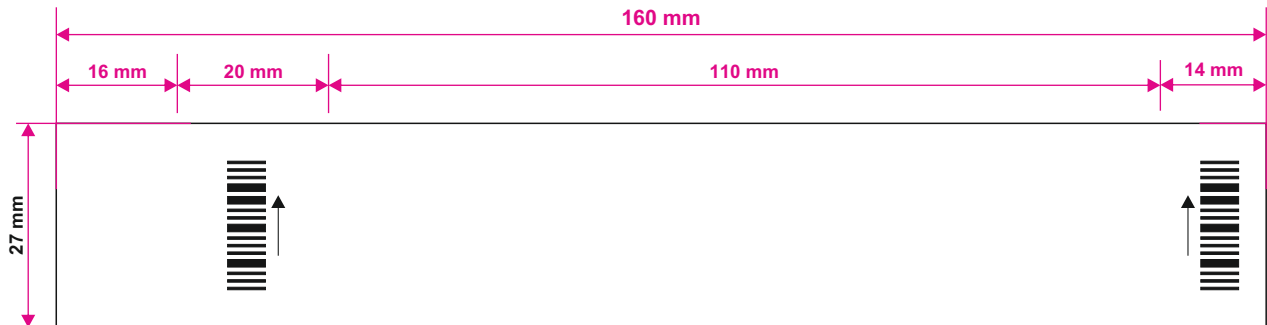




For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Nebimac 2.5 (Nebivolol Tablets 2.5 mg)

Nebimac 5 (Nebivolol Tablets 5 mg)

COMPOSITION

Each uncoated tablet contains:
Nebivolol Hydrochloride
equivalent to Nebivolol2.5 mg / 5 mg

PRODUCT DESCRIPTION

For 2.5 mg: White to off white, capsule shape, biconvex uncoated tablets debossed with "T 55" on one side and score line on the other side.

For 5 mg: White to off white, circular, biconvex uncoated tablets, debossed with "T 56" on one side and cross score line on the other side.

PHARMACOLOGICAL CLASSIFICATION

Selective beta-receptor antagonist

PHARMACOLOGICAL ACTION:

Pharmacodynamics

Selective Beta blocking agent, ATC code: C07AB12

Nebivolol is a race-mate of two enantiomers, SRRR-nebivolol (or d-nebivolol) and RSSS-nebivolol (or l-nebivolol). It combines two pharmacological activities: It is a competitive and selective beta-receptor antagonist: this effect is attributed to the SRRR-enantiomer (d-enantiomer). It has mild vasodilating properties, possibly due to an interaction with the L-arginine/nitric oxide pathway. Single and repeated doses of nebivolol reduce heart rate and blood pressure at rest and during exercise, both in normotensive subjects and in hypertensive patients. The antihypertensive effect is maintained during chronic treatment. At therapeutic doses, nebivolol is devoid of alpha-adrenergic antagonism. During acute and chronic treatment with nebivolol in hypertensive patients systemic vascular resistance is decreased. Despite heart rate reduction, reduction in cardiac output during rest and exercise may be limited due to an increase in stroke volume. The clinical relevance of these haemodynamic differences as compared to other beta1 receptor antagonists has not been fully established.

In hypertensive patients, nebivolol increases the NO-mediated vascular response to acetylcholine (ACh) which is reduced in patients with endothelial dysfunction. In a mortality-morbidity, placebo-controlled trial performed in 2128 patients ≥ 70 years (median age 75.2 years) with stable chronic heart failure with or without impaired left ventricular ejection fraction (mean LVEF: $36 \pm 12.3\%$, with the following distribution: LVEF less than 35% in 56% of patients, LVEF between 35% and 45% in 25% of patients and LVEF greater than 45% in 19% of patients) followed for a mean time of 20 months, nebivolol, on top of standard therapy, significantly prolonged the time to occurrence of deaths or hospitalisations for cardiovascular reasons (primary end-point for efficacy) with a relative risk reduction of 14% (absolute reduction: 4.2%). This risk reduction developed after 6 months of treatment and was maintained for all treatment duration (median duration: 18 months). The effect of nebivolol was independent from age, gender, or left ventricular ejection fraction of the population on study. The benefit on all-cause mortality did not reach statistical significance in comparison to placebo (absolute reduction: 2.3%). A decrease in sudden death was observed in nebivolol treated patients (4.1% vs 6.6%, relative reduction of 38%). In vitro and in vivo experiments in animals showed that Nebivolol has no intrinsic sympathomimetic activity. In vitro and in vivo experiments in animals showed that at pharmacological doses nebivolol has no membrane stabilising action. In healthy volunteers, nebivolol has no significant effect on maximal exercise capacity or endurance. Available preclinical and clinical evidence in hypertensive patients has not shown that nebivolol has a detrimental effect on erectile function.

Pharmacokinetics

Both nebivolol enantiomers are rapidly absorbed after oral administration. The absorption of nebivolol is not affected by food; nebivolol can be given with or without meals. Nebivolol is extensively metabolised, partly to active hydroxy-metabolites. Nebivolol is metabolised via allicyclic and aromatic hydroxylation, N-dealkylation and glucuronidation; in addition, glucuronides of the hydroxy-metabolites are formed. The metabolism of nebivolol by aromatic hydroxylation is subject to the CYP2D6 dependent genetic oxidative polymorphism. The oral bioavailability of nebivolol averages 12% in fast metabolisers and is virtually complete in slow metabolisers. At steady state and at the same dose level, the peak plasma concentration of unchanged nebivolol is about 23 times higher in poor metabolisers than in extensive metabolisers. When unchanged drug plus active metabolites are considered, the difference in peak plasma concentrations is 1.3 to 1.4 fold. Because of the variation in rates of metabolism, the dose of Nebivolol Tablets should always be adjusted to the individual requirements of the patient: poor metabolisers therefore may require lower doses. In fast metabolisers, elimination half-lives of the nebivolol enantiomers average 10 hours. In slow metabolisers, they are 3-5 times longer. In fast metabolisers, plasma levels of the RSSS-enantiomer are slightly higher than for the SRRR-enantiomer. In slow metabolisers, this difference is larger. In fast metabolisers, elimination half-lives of the hydroxy-metabolites of both enantiomers average 24 hours, and are about twice as long in slow metabolisers. Steady-state plasma levels in most subjects (fast metabolisers) are reached within 24 hours for nebivolol and within a few days for the hydroxy-metabolites. Plasma concentrations are dose-proportional between 1 and 30 mg. The pharmacokinetics of nebivolol is not affected by age.

In plasma, both nebivolol enantiomers are predominantly bound to albumin. Plasma protein binding is 98.1% for SRRR-nebivolol and 97.9% for RSSS-nebivolol. One week after administration, 38% of the dose is excreted in the urine and 48% in the faeces. Urinary excretion of unchanged nebivolol is less than 0.5% of the dose.

INDICATIONS AND USAGE

Hypertension

Treatment of essential hypertension

Chronic heart failure (CHF)

Treatment of stable mild and moderate chronic heart failure in addition to standard therapies in elderly patients ≥ 70 years

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients.
- Liver insufficiency or liver function impairment.
- Acute heart failure, cardiogenic shock or episodes of heart failure decompensation requiring i.v. inotropic therapy
- In addition, as with other beta-blocking agents, Nebivolol Tablets is contra-indicated in:
 - Sick sinus syndrome, including sino-atrial block.
 - Second and third degree heart block (without a pacemaker).
 - Brady cardia (heart rate < 60 bpm prior to start therapy).
 - Hypotension (systolic blood pressure < 90 mmHg).
 - History of bronchospasm and bronchial asthma.
 - Untreated phaeochromocytoma.
 - Metabolic acidosis.
 - Severe peripheral circulatory disturbances.

ADVERSE REACTION

Adverse events are listed separately for hypertension and CHF because of differences in the background diseases.

Hypertension

The adverse reactions reported, which are in most of the cases of mild to moderate intensity, are tabulated below, classified by system organ class and ordered by frequency:

System Organ Class	Common ($\geq 1/100$ to < 1/10)	Uncommon ($\geq 1/1,000$ to $\leq 1/100$)	Very Rare ($\leq 1/10,000$)	Not Known
Immune system disorders				angioneurotic oedema, hypersensitivity
Psychiatric disorders		nightmares; depression		angioneurotic oedema, hypersensitivity
Nervous system disorders	headache, dizziness, paraesthesia		syncope	
Eye disorders		impaired vision		
Cardiac disorders		bradycardia, heart failure, slowed AV conduction/AV-block		
Vascular disorders		hypotension, (increase of) intermittent claudication		
Respiratory, thoracic and mediastinal disorders	dyspnoea	bronchospasm		
Gastrointestinal disorders	constipation, nausea, diarrhoea	dyspepsia, flatulence, vomiting		
Skin and subcutaneous tissue disorders		pruritus, rash erythematous	psoriasis aggravated	urticaria
Reproductive system and breast disorders		impotence		
General disorders and administration site conditions	tiredness,			

The following adverse events have also been reported with some beta adrenergic antagonists: hallucinations, psychoses, confusion, cold/cyanotic extremities, Raynaud phenomenon, dry eyes, and oculo-mucocutaneous toxicity of the practolol-type.

Chronic heart failure

Data on adverse reactions in CHF patients are available from one placebo-controlled clinical trial involving 1061 patients taking nebivolol and 1061 patients taking placebo. In this study, a total of 449 nebivolol patients (42.1%) reported at least possibly causally related adverse reactions compared to 334 placebo patients (31.5%). The most commonly reported adverse reactions in nebivolol patients were bradycardia and dizziness, both occurring in approximately 11% of patients. The corresponding frequencies among placebo patients were approximately 2% and 7%, respectively.

The following incidences were reported for adverse reactions (at least possibly drug-related) which are considered specifically relevant in the treatment of chronic heart failure:

- Aggravation of cardiac failure occurred in 5.8% of nebivolol patients compared to 5.2% of placebo patients.
- Postural hypotension was reported in 2.1% of nebivolol patients compared to 1.0% of placebo patients.
- Drug intolerance occurred in 1.6% of nebivolol patients compared to 0.8% of placebo patients.
- First degree atrio-ventricular block occurred in 1.4% of nebivolol patients compared to 0.9% of placebo patients.
- Oedema of the lower limb was reported by 1.0% of nebivolol patients compared to 0.2% of placebo patients.

DOSAGE

Hypertension

Adults

For 2.5 mg: The dose is 5 mg (two 2.5mg tablets) daily, preferably at the same time of the day. Tablets may be taken with meals.

The blood pressure lowering effect becomes evident after 1-2 weeks of treatment. Occasionally, the optimal effect is reached only after 4 weeks.

For 5 mg: The dose is one tablet (5 mg) daily, preferably at the same time of the day. The blood pressure lowering effect becomes evident after 1-2 weeks of treatment. Occasionally, the optimal effect is reached only after 4 weeks.

Combination with other antihypertensive agents

Beta-blockers can be used alone or concomitantly with other antihypertensive agents. Till date, an additional antihypertensive effect has been observed only when Nebivolol Tablets is combined with hydrochlorothiazide 12.5-25 mg.

Patients with renal insufficiency

In patients with renal insufficiency, the recommended starting dose is 2.5 mg daily. If needed, the daily dose may be increased to 5 mg.

Patients with hepatic insufficiency

Data in patients with hepatic insufficiency or impaired liver function are limited. Therefore the use of Nebivolol Tablets 2.5mg or Nebivolol Tablets 5mg in these patients is contraindicated.

Older people

In patients over 65 years, the recommended starting dose is 2.5 mg daily. If needed, the daily dose may be increased to 5 mg. However, in view of the limited experience in patients above 75 years, caution must be exercised and these patients monitored closely.

Paediatric population

The efficacy and safety of Nebivolol Tablets 2.5 mg & 5mg in children and adolescents aged below 18 years has not been established. No data are available. Therefore, use in children is not recommended.

Chronic heart failure (CHF)

The treatment of stable chronic heart failure has to be initiated with a gradual up-titration of dosage until the optimal individual maintenance dose is reached. Patients should have stable chronic heart failure without acute failure during the past six weeks. It is recommended that the treating physician should be experienced in the management of chronic heart failure. For those patients receiving cardiovascular drug therapy including diuretics and/or digoxin and/or ACE inhibitors and/or angiotensin II antagonists, dosing of these drugs should be stabilised during the past two weeks prior to initiation of Nebivolol treatment.

The initial up-titration should be done according to the following steps at 1-2 weekly intervals based on patient tolerability:

1.25 mg nebivolol, to be increased to 2.5 mg nebivolol once daily, then to 5 mg once daily and then to 10 mg once daily. The maximum recommended dose is 10 mg nebivolol once daily.

240 mm

160 mm

For a correct breaking of the tablet the below instructions have to be followed;

- Tablet has to be placed onto a flat, hard surface (e.g. a table or worktop), with the cross score facing up.
- Break the tablet by pushing it with the index fingers of both hands placed along one breakmark (Diagrams 1 and 2).



Diagrams 1 and 2: Easy breaking of the Nebivolol Tablets cross-scored tablet in half



Diagrams 3 and 4: Easy breaking of half of the Nebivolol tablets cross-scored tablet into quarters

Initiation of therapy and every dose increase should be done under the supervision of an experienced physician over a period of at least 2 hours to ensure that the clinical status (especially as regards blood pressure, heart rate, conduction disturbances, signs of worsening of heart failure) remains stable. Occurrence of adverse events may prevent all patients being treated with the maximum recommended dose. If necessary, the dose reached can also be decreased step by step and reintroduced as appropriate. During the titration phase, in case of worsening of the heart failure or intolerance, it is recommended first to reduce the dose of nebivolol, or to stop it immediately if necessary (in case of severe hypotension, worsening of heart failure with acute pulmonary oedema, cardiogenic shock, symptomatic bradycardia or AV block). Treatment of stable chronic heart failure with nebivolol is generally a long-term treatment. The treatment with nebivolol is not recommended to be stopped abruptly since this might lead to a transitory worsening of heart failure. If discontinuation is necessary, the dose should be gradually decreased divided into halves weekly.

Patients with renal insufficiency

No dose adjustment is required in mild to moderate renal insufficiency since up-titration to the maximum tolerated dose is individually adjusted. There is no experience in patients with severe renal insufficiency (serum creatinine $\geq 250 \mu\text{mol/L}$). Therefore, the use of nebivolol in these patients is not recommended.

Patients with hepatic insufficiency

Data in patients with hepatic insufficiency are limited. Therefore the use of Nebivolol Tablets in these patients is contra-indicated.

Elderly People

No dose adjustment is required since up-titration to the maximum tolerated dose is individually adjusted.

Paediatric population

The efficacy and safety of Nebivolol Tablets in children and adolescents aged below 18 years has not been established. No data are available. Therefore, use in children and adolescents is not recommended.

Method of administration

Oral use

Tablets may be taken with meals.

DRUG INTERACTIONS

The following interactions apply to beta-adrenergic antagonists in general.

Calcium antagonists: Care should be exercised when administering beta-adrenergic antagonists with calcium antagonists of the verapamil or diltiazem type, because of their negative effect on contractility and atrio-ventricular conduction. Intravenous verapamil is contra-indicated in patients on Nebivolol Tablets.

Anti-arrhythmics: Caution should be exercised when administering beta-adrenergic antagonists in association with Class I anti-arrhythmic drugs and amiodarone, as their effect on atrial conduction time and their negative inotropic effect may be potentiated.

Centrally-acting antihypertensives (clonidine, guanfacin, moxonidine, methyl dopa, rilmenidine): Concomitant use of centrally acting antihypertensive drugs may worsen heart failure by a decrease in the central sympathetic tone (reduction of heart rate and cardiac output, vasodilation). Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of "rebound hypertension".

Digoxin: Digitalis glycosides associated with beta-adrenergic antagonists may increase atrioventricular conduction time. Clinical trials with nebivolol have not shown any clinical evidence of an interaction. Nebivolol does not influence the kinetics of digoxin.

Insulin and oral antidiabetic drugs: Although Nebivolol Tablets does not affect glucose levels; certain symptoms of hypoglycaemia (palpitations, tachycardia) may be masked.

Anaesthetics: Concomitant use of beta-adrenergic antagonists and anaesthetics may attenuate reflex tachycardia and increase the risk of hypotension. As a general rule, avoid sudden withdrawal of beta-blocker treatment. The anaesthesiologist should be informed when the patient is receiving Nebivolol Tablets.

Other: Class III antiarrhythmic drugs (Amiodarone): effect on atrio-ventricular conduction time may be potentiated

Baclofen (antispastic agent), amifostine (antineoplastic adjunct): concomitant use with antihypertensives is likely to increase the fall in blood pressure; therefore the dosage of the antihypertensive medication should be adjusted accordingly

Calcium antagonists of the dihydropyridine type (amlodipine, felodipine, lacidipine, nifedipine, nicardipine, nimodipine, nitrendipine): concomitant use may increase the risk of hypotension, and an increase in the risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.

Concomitant use of NSAIDs had no effect on the blood pressure lowering effect of Nebivolol Tablets. Co-administration of cimetidine increased the plasma levels of nebivolol, without changing the clinical effect. Co-administration of ranitidine did not affect the pharmacokinetics of nebivolol.

Provided Nebivolol Tablet is taken with the meal, and an antacid between meals, the two treatments can be co-prescribed. Combining nebivolol with nicardipine slightly increased the plasma levels of both drugs, without changing the clinical effect.

Co-administration of alcohol, furosemide or hydrochlorothiazide did not affect the pharmacokinetics of nebivolol. Nebivolol does not affect the pharmacokinetics and pharmacodynamics of warfarin.

Sympathomimetic agents may counteract the effect of beta-adrenergic antagonists. Beta-adrenergic agents may lead to unopposed alpha-adrenergic activity of sympathomimetic agents with both alpha- and beta-adrenergic effects (risk of hypertension, severe bradycardia and heart block). Concomitant administration of tricyclic antidepressants, barbiturates and phenothiazines may increase the blood pressure lowering effect. As nebivolol metabolism involves the CYP2D6 isoenzyme, concomitant administration of serotonin reuptake inhibitors, dextromethorphan or other compounds predominantly metabolised via this pathway, may make extensive metabolisers resemble poor metabolisers.

WARNING AND PRECAUTIONS

The following warnings and precautions apply to beta-adrenergic antagonists in general.

Anaesthesia:

Continuation of beta blockade reduces the risk of arrhythmias during induction and intubation. If beta blockade is interrupted in preparation for surgery, the beta-adrenergic antagonist should be discontinued at least 24 hours beforehand. Caution should be observed with certain anaesthetics that cause myocardial depression, such as cyclopropane, ether or trichloroethylene. The patient can be protected against vagal reactions by intravenous administration of atropine.

Cardiovascular

In general, beta-adrenergic antagonists should not be used in patients with untreated congestive heart failure (CHF), unless their condition has been stabilised. In patients with ischaemic heart disease, treatment with a beta-adrenergic antagonist should be discontinued gradually, i.e. over 1-2 weeks. If necessary replacement therapy should be initiated at the same time to prevent exacerbation of angina pectoris. Beta-adrenergic antagonists may induce bradycardia: if the pulse rate drops below 50-55 bpm at rest and/or the patient experiences symptoms are suggestive of bradycardia, the dosage should be reduced. Beta-adrenergic antagonists should be used with caution: in patients with peripheral circulatory disorders (Raynaud's disease or syndrome, intermittent claudication), as aggravation of these disorders may occur; in patients with first degree heart block, because of the negative effect of beta-blockers on conduction time; in patients with Prinzmetal's angina due to unopposed alpha-receptor mediated coronary artery vasoconstriction: beta-adrenergic antagonists may increase the number and duration of anginal attacks. Combination of nebivolol with calcium channel antagonists of the verapamil and diltiazem type, with Class I antiarrhythmic drugs, and with centrally acting antihypertensive drugs is generally not recommended.

Metabolic/Endocrinological

Nebivolol Tablets does not affect glucose levels in diabetic patients. Care should be taken in diabetic patients however, as nebivolol may mask certain symptoms of hypoglycaemia (tachycardia, palpitations). Beta- adrenergic blocking agents may mask tachycardic symptoms in hyperthyroidism. Abrupt withdrawal may intensify symptoms.

Respiratory

In patients with chronic obstructive pulmonary disorders, beta-adrenergic antagonists should be used with caution as airway constriction may be aggravated.

Other

Patients with a history of psoriasis should take beta-adrenergic antagonists only after careful consideration. Beta-adrenergic antagonists may increase the sensitivity to allergens and the severity of anaphylactic reactions.

Use in children is not recommended.

The initiation of Chronic Heart Failure treatment with nebivolol necessitates regular monitoring. Treatment discontinuation should not be done abruptly unless clearly indicated. This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp-lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

PREGNANCY AND LACTATION:

Pregnancy

Nebivolol has pharmacological effects that may cause harmful effects on pregnancy and/or the foetus/new-born. In general, beta-adrenoceptor blockers reduce placental perfusion, which has been associated with growth retardation, intrauterine death, abortion or early labour. In addition, adverse effects (hypoglycaemia and bradycardia) may occur in the fetus and the neonate. If treatment with beta-adrenoceptor blockers is necessary, beta1-selective adrenoceptor blockers are preferable. Nebivolol should not be used during pregnancy unless clearly necessary. If treatment with nebivolol is considered necessary, the utero placental blood flow and the foetal growth should be monitored. In case of harmful effects on pregnancy or the foetus alternative treatment should be considered. The new-born infant must be closely monitored. Symptoms of hypoglycaemia and bradycardia are generally to be expected within the first 3 days

Breast-feeding

Most beta-blockers, particularly lipophilic compounds like nebivolol and its active metabolites, pass into breast milk although to a variable extent. Since it is not known whether nebivolol is excreted into human milk; the use of Nebivolol Tablets when breast feeding is contra-indicated. Animal studies have shown that nebivolol is excreted in breast milk.

Effects on ability to drive and use machines

There are no studies on the effect of Nebivolol Tablets on the ability to drive. Pharmacodynamic studies have shown that Nebivolol Tablets does not affect psychomotor function. When driving vehicles or operating machines it should be taken into account that dizziness and fatigue may occasionally occur.

OVERDOSAGE

No data are available on over dosage with Nebivolol Tablets.

Symptoms:

Symptoms of over dosage with beta-blockers are: bradycardia, hypotension, bronchospasm and acute cardiac insufficiency.

Treatment:

In case of over dosage or hypersensitivity, the patient should be kept under close supervision and be treated in an intensive care ward. Blood glucose levels should be checked. Absorption of any drug residues still present in the gastro-intestinal tract can be prevented by gastric lavage and the administration of activated charcoal and a laxative. Artificial respiration may be required. Bradycardia or extensive vagal reactions should be treated by administering atropine or methylatropine. Hypotension and shock should be treated with plasma/plasma substitutes and, if necessary, catecholamines. The beta- blocking effect can be counteracted by slow intravenous administration of isoprenaline hydrochloride, starting with a dose of approximately 5 mg/minute, or dobutamine, starting with a dose of 2.5 mg/minute, until the required effect has been obtained. In refractory cases isoprenaline can be combined with dopamine. If this does not produce the desired effect either, intravenous administration of glucagon 50-100 mg/kg i.v. may be considered. If required, the injection should be repeated within one hour, to be followed -if required- by an i.v. infusion of glucagon 70 mg/kg/h. In extreme cases of treatment-resistant bradycardia, a pacemaker may be inserted.

STORAGE

Store below 30°C in dry place

PRESENTATION

For 2.5 mg & 5mg: Alu/Clear PVC/PVDC blister pack of 10 Tablets
Available in pack size of 3 x 10 Tablets

MANUFACTURED BY

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DATE OF REVISION:

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PM04311801

240 mm